

CHAPTER 1

PHYSICAL QUANTITIES, UNITS & MEASUREMENTS

1.1 Physical quantities and measurement techniques

- 1 Describe how to measure a variety of lengths with appropriate precision using tapes, rulers and micrometers (including reading the scale on an analogue micrometer)
- 2 Describe how to use a measuring cylinder to measure the volume of a liquid and to determine the volume of a solid by displacement
- 3 Describe how to measure a variety of time intervals using clocks and digital timers
- 4 Determine an average value for a small distance and for a short interval of time by measuring multiples (including the period of oscillation of a pendulum)
- 5 Understand that a scalar quantity has magnitude (size) only and that a vector quantity has magnitude and direction
- 6 Know that the following quantities are scalars: distance, speed, time, mass, energy and temperature
- 7 Know that the following quantities are vectors: displacement, force, weight, velocity, acceleration, momentum, electric field strength and gravitational field strength
- 8 Determine, by calculation or graphically, the resultant of two vectors at right angles

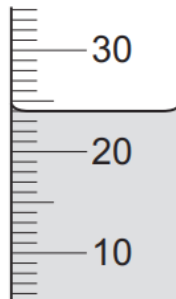
1. O/N/23/11/Q1

Which description is of a scalar quantity?

- A** It has a direction only.
- B** It has direction and unit only.
- C** It has direction, magnitude and unit.
- D** It has magnitude and unit only.

2. O/N/23/12/Q1

The diagram shows the level of liquid in a measuring cylinder. The scale is marked in cm^3 .



What is the volume of the liquid?

- A** 20.4 cm^3 **B** 24.0 cm^3 **C** 24.5 cm^3 **D** 25.0 cm^3

3. M/J/23/11/Q1

A student wishes to measure directly the circumference of a football.

What is the most suitable instrument to use?

- A** a clock only
- B** a measuring tape only
- C** a micrometer only
- D** a ruler only

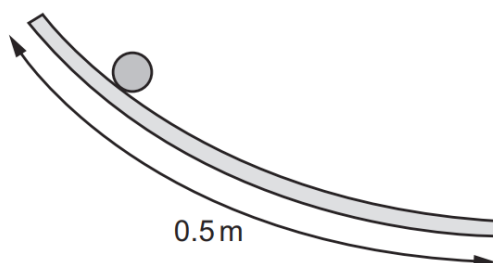
4. M/J/23/11/Q2

Which equation contains two vector quantities?

- A** $\text{acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$
- B** $\text{average speed} = \frac{\text{distance travelled}}{\text{time taken}}$
- C** $\text{density} = \frac{\text{mass}}{\text{volume}}$
- D** $\text{volume} = \text{length} \times \text{width} \times \text{height}$

5. M/J/23/12/Q1

In an experiment, a ball is rolled down a curved track that is about half a metre long.



Which measuring device is used to measure the length accurately?

- A** metre rule
- B** micrometer
- C** stop-watch
- D** tape measure

6. M/J/23/12/Q2

Which row describes acceleration, displacement, distance and speed?

	acceleration	displacement	distance	speed
A	scalar	scalar	vector	scalar
B	scalar	vector	scalar	vector
C	vector	scalar	vector	vector
D	vector	vector	scalar	scalar

7. O/N/22/11/Q2

Velocity is given by the change in displacement divided by the change in time.

How many vector quantities appear in this statement?

- A** 0
- B** 1
- C** 2
- D** 3

8. O/N/22/12/Q3

Which value is one-thousandth of a metre?

- A** 0.0001 cm
- B** 0.001 cm
- C** 0.01 cm
- D** 0.1 cm

9. O/N/21/11/Q1

Which quantity is a vector?

- A** speed
- B** force
- C** mass
- D** distance

10. O/N/21/12/Q2

A student investigates the motion of a ball falling through the air.

Which quantity is a vector?

- A** the diameter of the ball
- B** the gravitational force on the ball
- C** the distance from which the ball is dropped
- D** the speed at which the ball hits the ground

11. M/J/21/11/Q1

A list of various quantities is shown.

acceleration

displacement

force

length

mass

velocity

How many of these quantities are vectors?

- A** 2 **B** 3 **C** 4 **D** 5

12. M/J/20/11/Q2

What is measured using a micrometer?

- A** area
- B** current
- C** length
- D** mass

13. O/N/19/11/Q1

Which word is the name of a scalar quantity?

- A** acceleration
- B** force
- C** mass
- D** velocity

14. O/N/19/11/Q4

What is the correct unit for the quantity shown?

	quantity	unit
A	electromotive force (e.m.f.)	N
B	latent heat	J
C	pressure	kg / m ³
D	weight	kg

15. O/N/19/12/Q1

Which word is the name of a vector quantity?

- A** density
- B** displacement
- C** energy
- D** speed

16. M/J/19/11/Q1

Which quantities are both vectors?

- A** acceleration and length
- B** distance and velocity
- C** length and distance
- D** velocity and acceleration

17. M/J/19/11/Q4

The magnitudes of three different electric charges are given below.

What is the correct order of size, from largest to smallest?

- A** 1 mC → 1 MC → 1 kC
- B** 1 MC → 1 mC → 1 kC
- C** 1 MC → 1 kC → 1 mC
- D** 1 kC → 1 mC → 1 MC

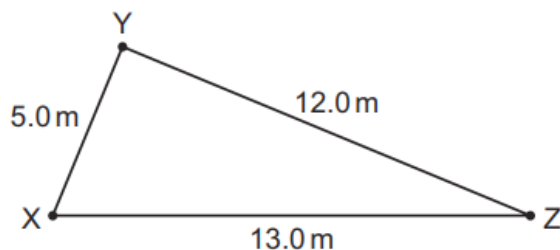
18. M/J/19/12/Q1

Which reading is given to one tenth of a millimetre?

- A** 3.3 cm **B** 3.31 cm **C** 3.310 cm **D** 3.312 cm

19. O/N/18/12/Q9

Paths are laid as shown between points X, Y and Z.



A person walks along the paths from X to Y to Z and then back to X.

What is the value of the total displacement and of the total distance travelled?

	total displacement /m	total distance travelled /m
A	0	0
B	0	30
C	30	0
D	30	30

20. O/N/18/11/Q1

What is a possible mass for a normal adult person?

- A** 7.5 kg **B** 75 kg **C** 750 kg **D** 7500 kg

21. O/N/17/11/Q1

Which list contains only scalar quantities?

- A** acceleration, displacement, velocity
B distance, force, speed
C force, length, time
D length, mass, speed

22. M/J/16/11/Q2

Which set of quantities are all vectors?

- A** acceleration, displacement, velocity
- B** chemical energy, mass, power
- C** extension, force, gravitational potential energy
- D** weight, kinetic energy, work

23. M/J/15/12/Q1

Which quantity is a scalar?

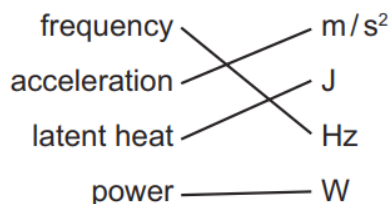
- A** acceleration
- B** force
- C** temperature
- D** velocity

24. O/N/14/11/Q3

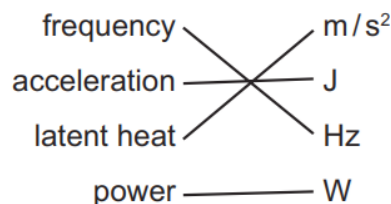
In a test, four students linked the quantities on the left with their units on the right.

Which student matched them all correctly?

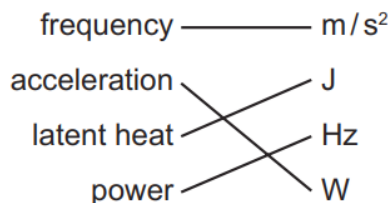
A



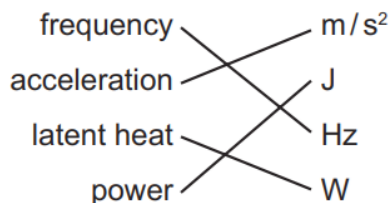
B



C



D



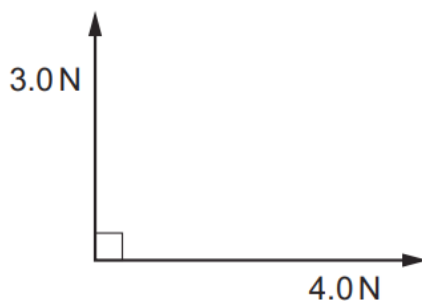
25. O/N/14/11/Q2

Which device can be used to measure the thickness of a single sheet of paper?

- A** a metre rule
- B** a micrometer
- C** a plastic ruler
- D** a measuring tape

26. M/J/14/12/Q2

What is the size of the resultant of the two forces shown in the diagram?



- A** 1.0 N **B** 3.5 N **C** 5.0 N **D** 7.0 N

27. M/J/14/12/Q1

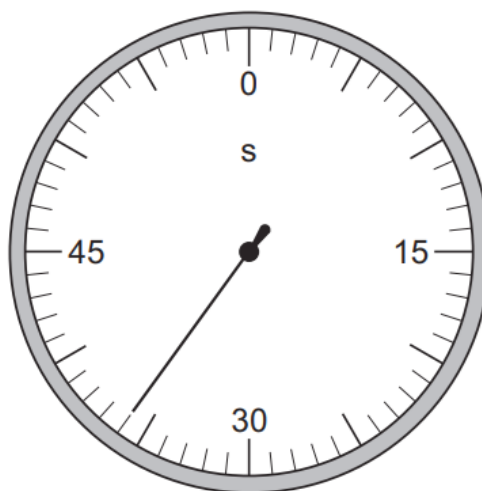
Each row contains a vector and a scalar.

In which row is the size of the vector equal to the size of the scalar?

	vector	scalar
A	displacement of a car	speed of the car
B	velocity of a car	distance travelled by the car
C	velocity of a car	speed of the car
D	weight of a car	mass of the car

28. M/J/14/11/Q2

The diagram shows a stopwatch.



What is the reading on the stopwatch?

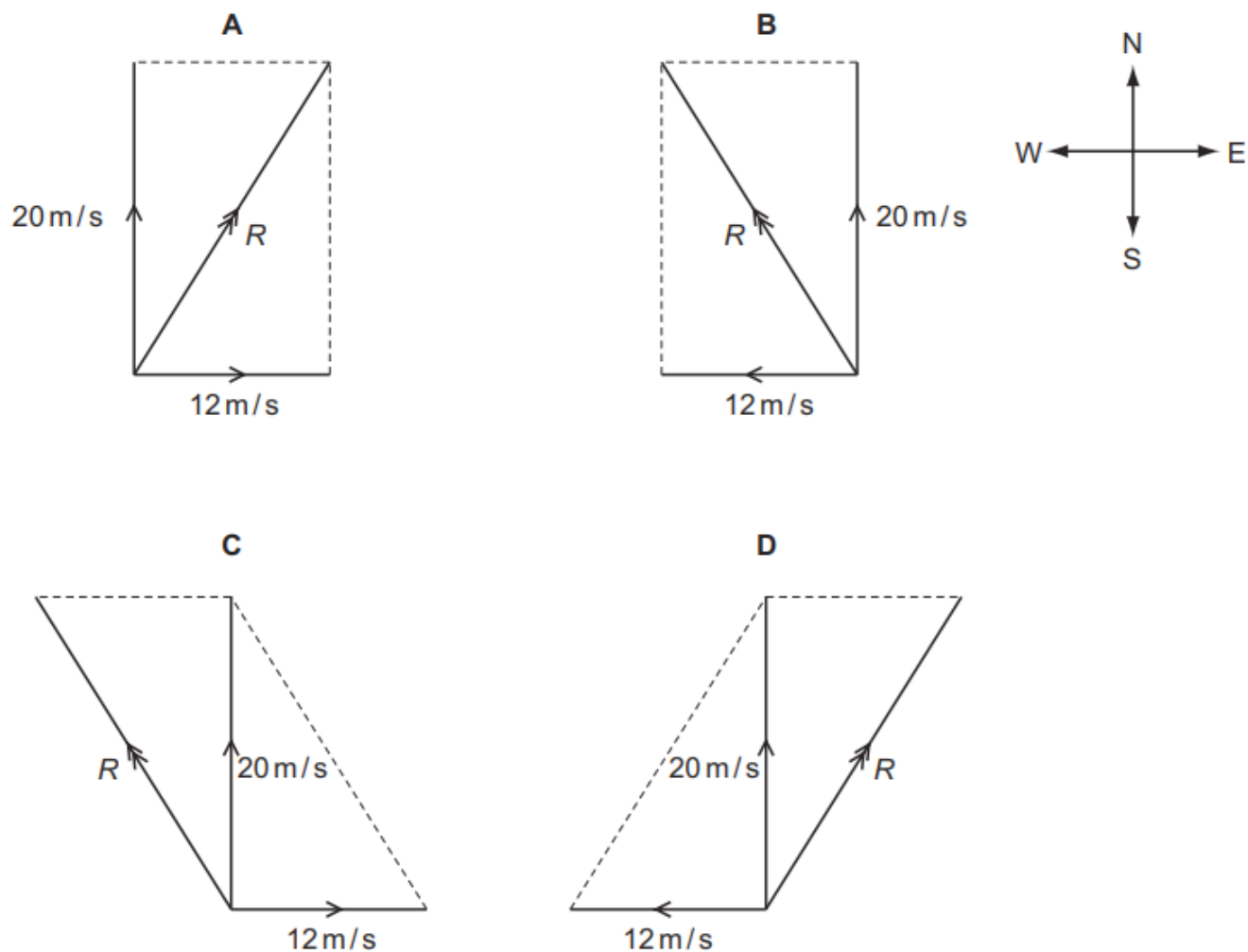
- A** 30.6 s **B** 33.0 s **C** 36.0 s **D** 36.6 s

29. O/N/13/11/Q1

When there is no wind, the engines of an airship push it due north at 20 m/s.

The wind is blowing from the west at 12 m/s.

Which vector diagram correctly shows how the resultant velocity R of the airship is obtained?



30. M/J/13/12/Q2

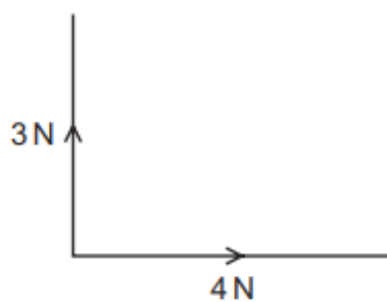
Before marking the finishing line on a running track, a groundsman measures out its 100 m length.

Which instrument is the most appropriate for this purpose?

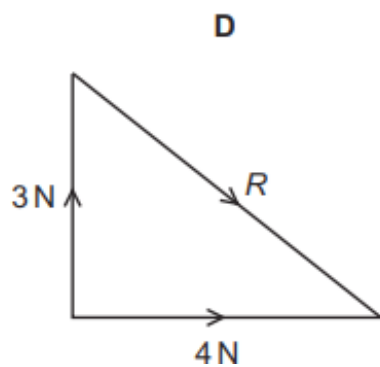
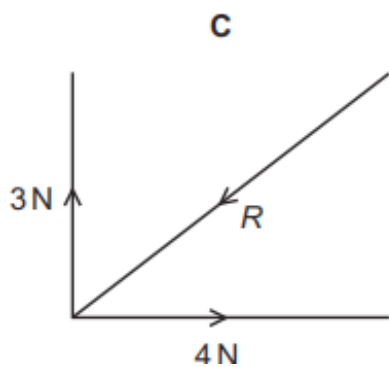
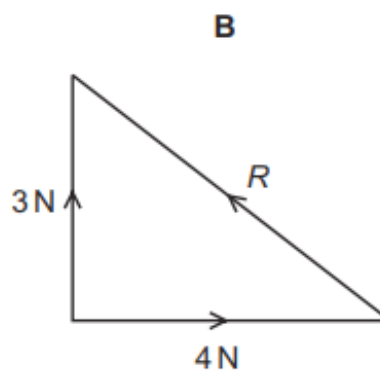
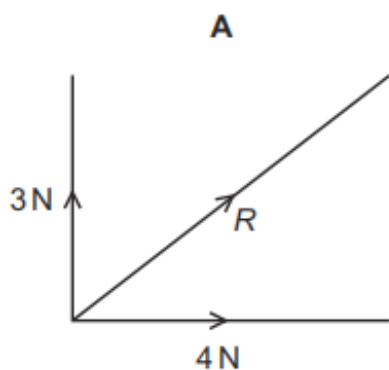
- | | |
|-------------------------|----------------------|
| A measuring tape | C 30 cm ruler |
| B metre rule | D micrometer |

31. M/J/13/11/Q1

Forces of 3 N and 4 N act as shown in the diagram.

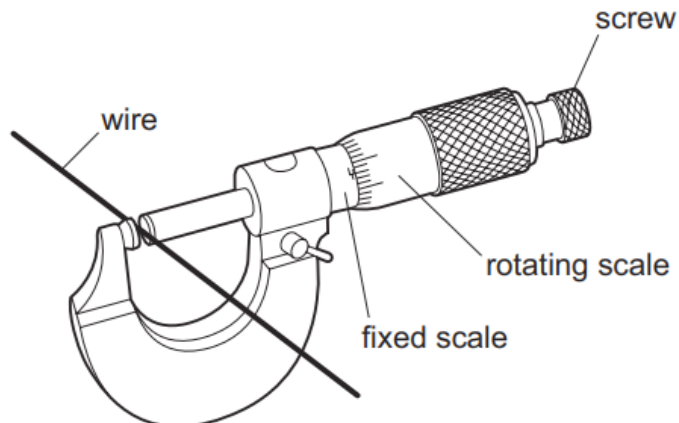


Which diagram shows the resultant R of these two forces?



32. M/J/13/11/Q2

A micrometer is used to measure the diameter of a uniform wire.



What is done to obtain an accurate answer?

- A** Find the reading and add or subtract the zero error.
- B** Make the micrometer horizontal.
- C** Subtract the fixed scale reading from the rotating scale reading.
- D** Subtract the rotating scale reading from the fixed scale reading.

33. M/J/12/12/Q2

Which row correctly shows examples of a vector quantity and a scalar quantity?

	vector	scalar
A	area	force
B	mass	density
C	velocity	acceleration
D	weight	volume

34. O/N/11/12/Q7

A quantity is quoted as having a value of 6.2 ms.

In what units is it measured?

- | | |
|----------------------------|-----------------------|
| A metres | C microseconds |
| B metres per second | D milliseconds |

35. M/J/11/12/Q1

What is the correct unit for the quantity shown?

	quantity	unit
A	electromotive force (e.m.f.)	N
B	latent heat	J
C	pressure	kg / m ³
D	weight	kg

36. M/J/10/11/Q2

The following statements are about motion.

- 1 A plane flies due East for 600 km.
- 2 A runner's average speed in a race around a track is 5 m/s.
- 3 A snail crawls at 3 mm/s in a straight line towards a lettuce.
- 4 A tourist travels 500 km on a journey.

Which statements describe vector quantities?

- A** 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 2 and 4

37. M/J/10/11/Q1

Power is measured in watts.

What is the correct symbol for millions of watts?

- A** mw **B** mW **C** Mw **D** MW

38. M/J/09/Q2

Which statement about scalars and vectors is correct?

- A** A scalar has direction but no size. **C** A vector has direction but no size.
B A scalar has size but no direction. **D** A vector has size but no direction.

39. O/N/06/Q2

A student studies some equations.

$$\text{power} = \text{work} / \text{time}$$

$$\text{force} = \text{mass} \times \text{acceleration}$$

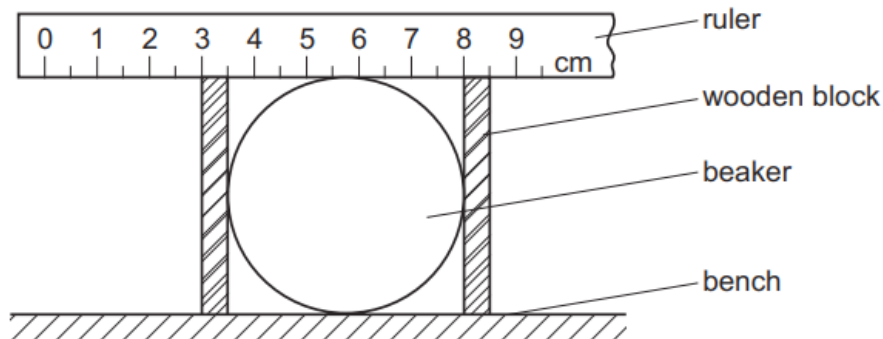
$$\text{velocity} = \text{displacement} / \text{time}$$

How many vector quantities are contained in the equations?

- A** 1 **B** 2 **C** 3 **D** 4

40. O/N/06/Q1

The diagram shows one method of measuring the diameter of a beaker.

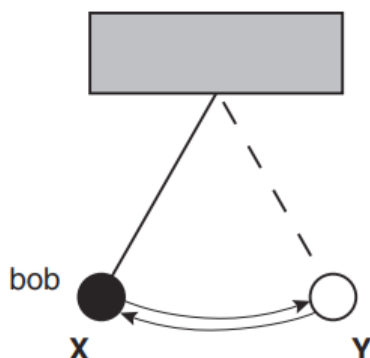


What is the diameter of the beaker?

- A** 4.5 cm **B** 5.0 cm **C** 5.5 cm **D** 8.0 cm

41. O/N/03/Q1

One oscillation of a swinging pendulum occurs when the bob moves from **X** to **Y** and back to **X** again.



Using a stopwatch, which would be the most accurate way to measure the time for one oscillation of the pendulum?

- A** Time 20 oscillations and multiply by 20.
B Time 20 oscillations and divide by 20.
C Time one oscillation.
D Time the motion from **X** to **Y**, and double it.

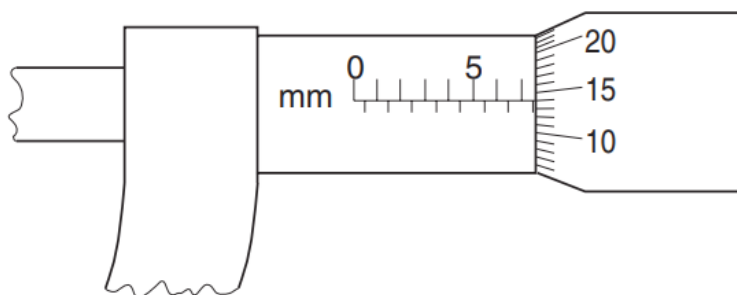
42. M/J/03/Q8

Which of the following groups of physical quantities consists only of scalars?

- | | |
|--|--------------------------------|
| A acceleration, force, velocity | C force, time, velocity |
| B acceleration, mass, speed | D mass, speed, time |

43. O/N/02/Q1

The diagram shows a micrometer scale.

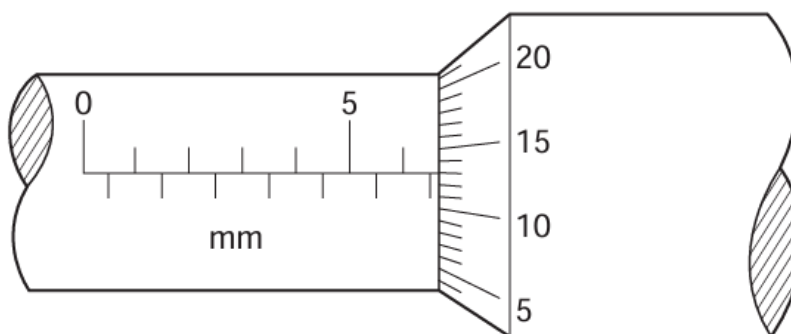


Which reading is shown?

- A** 5.64 mm **B** 7.14 mm **C** 7.16 mm **D** 7.64 mm

44. M/J/08/Q1

What is the reading on this micrometer?



- A** 5.43 mm **B** 6.63 mm **C** 7.30 mm **D** 8.13 mm

ANSWERS

- | | | |
|------|------|-------|
| 1. D | 16.D | 31. A |
| 2. B | 17.C | 32. A |
| 3. B | 18.B | 33. D |
| 4. A | 19.B | 34. D |
| 5. D | 20.B | 35. B |
| 6. D | 21.D | 36. B |
| 7. C | 22.A | 37. D |
| 8. D | 23.C | 38. B |
| 9. B | 24.A | 39. D |
| 10.B | 25.B | 40. A |
| 11.C | 26.C | 41. B |
| 12.C | 27.C | 42. D |
| 13.C | 28.C | 43. D |
| 14.B | 29.A | 44. B |
| 15.B | 30.A | |

1. O/N/23/22/Q1

Velocity is a vector quantity but speed is a scalar quantity.

- (a) State how a vector quantity differs from a scalar quantity.

.....
..... [1]

- (b) Underline **all** the vector quantities in the list.

acceleration **distance** **energy** **mass** **momentum** **temperature** [1]

- (c) Fig. 1.1 shows a passenger running across a large ship at right angles to the direction in which the ship is moving.

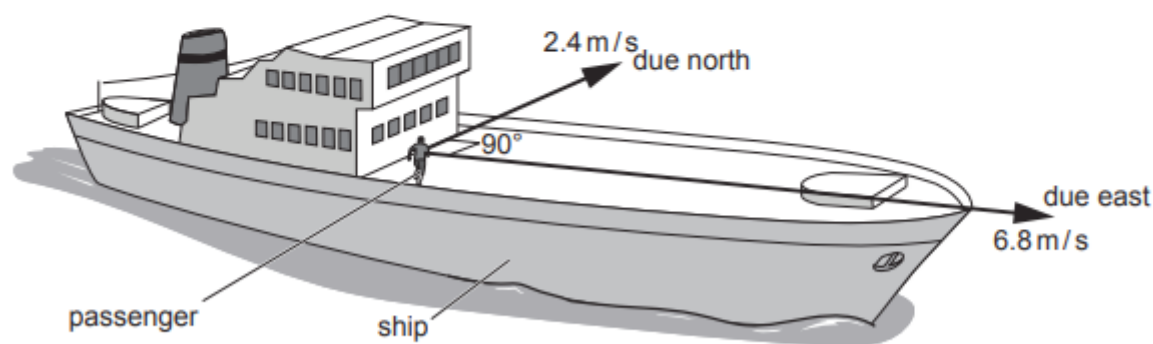


Fig. 1.1 (not to scale)

The passenger runs across the ship at 2.4 m/s due north.

The large ship is travelling due east at 6.8 m/s.

Determine, either graphically or by calculation, the resultant velocity of the passenger. Give the direction in which the passenger is moving relative to due north.

Show how the answer is obtained.

velocity = m/s at ° from due north [4]

[Total: 6]

2. M/J/22/21/Q7

Fig. 7.1 shows a toy helicopter. It can hover and travel through the air.

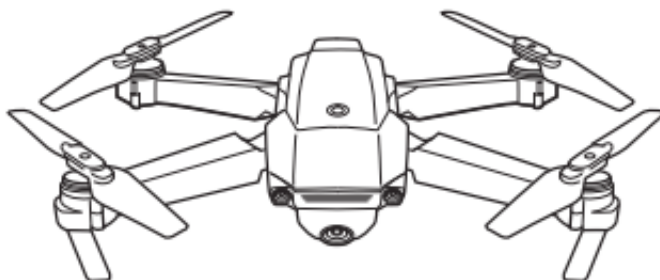


Fig. 7.1

A student flies the toy helicopter on a journey from A to B to C to D at a constant height.

Fig. 7.2 is a scale drawing of the path of the helicopter, viewed from above.

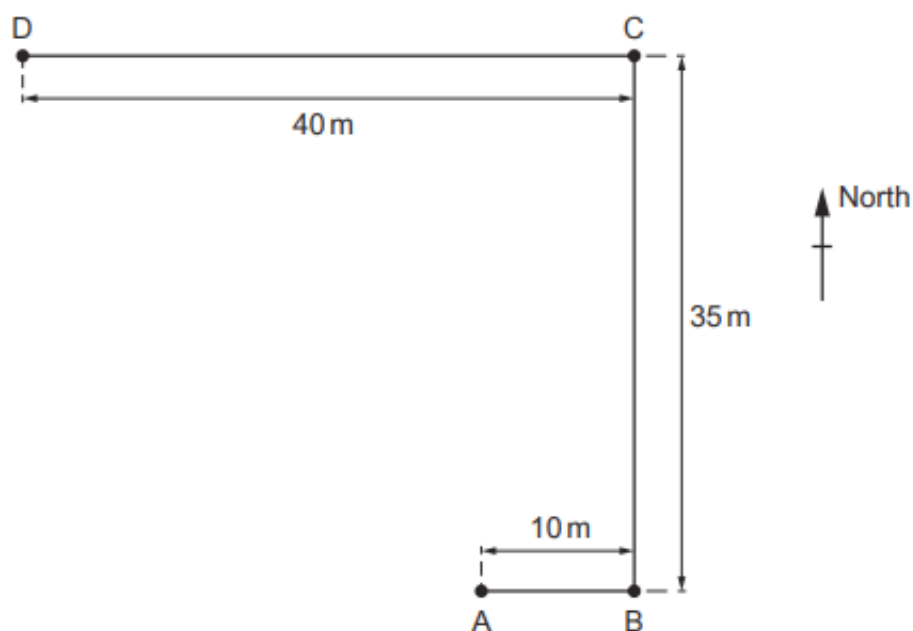


Fig. 7.2 (to scale)

- (a) (i) Determine the total distance travelled by the toy helicopter.

..... [1]

- (ii) The toy helicopter makes the journey in 40 s.

Calculate its average speed during the journey.

speed = [2]

(b) (i) State the difference between distance and displacement.

.....
..... [1]

(ii) Determine the scale used to construct Fig. 7.2 and complete the sentence.

1 cm on the diagram represents on the path. [1]

(iii) Using Fig. 7.2, determine the size of the displacement of point D from point A.

Show your working.

size of displacement = [2]

(iv) Determine the angle between North and the direction of the displacement of point D from point A.

angle = [1]

3. M/J/21/21/Q1

An aircraft flies at a constant height.

Air drag and the force from the aircraft's engines together produce a force on the aircraft of 36 kN due north, as shown in Fig. 1.1.

The wind produces a force of 12 kN towards the east.

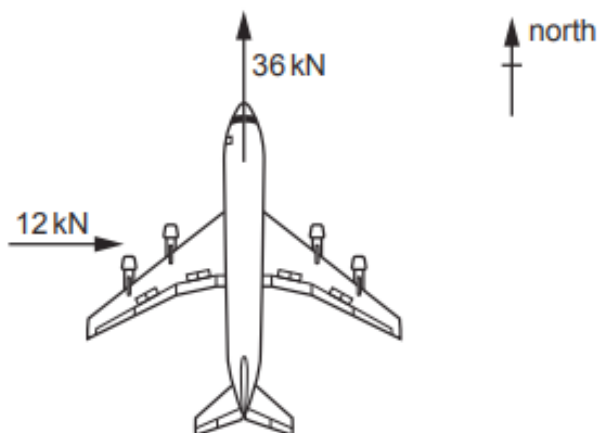


Fig. 1.1 (not to scale)

- (a) Draw a scale drawing to show the resultant force acting on the aircraft.

Use your drawing to determine the size of the resultant force and the angle between the resultant force and north.

size of resultant force =

angle =

[3]

4. M/J/21/22/Q1

(a) (i) State the difference between a scalar quantity and a vector quantity.

.....

.....

..... [1]

(ii) State **two** examples of each type of quantity.

scalar quantity	vector quantity
1	1
2	2

[2]

(b) Fig. 1.1 shows the direction and size of two vectors P and Q.

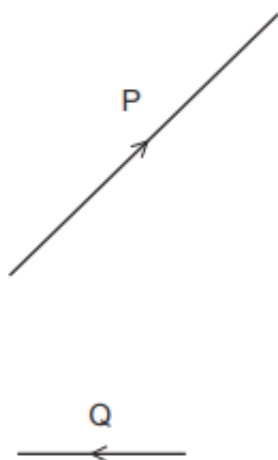


Fig. 1.1

In the space next to Fig. 1.1, draw a labelled vector diagram to show the resultant vector obtained by adding vector P to vector Q.

Draw vector P, vector Q and the resultant vector to the same scale as in Fig. 1.1.

[2]

[Total: 5]

5. O/N/15/21/Q9

Force is a vector quantity and mass is a scalar quantity.

- (a) (i)** State how a vector quantity differs from a scalar quantity.

.....
..... [1]

- (ii)** State one other vector quantity and one other scalar quantity.

vector quantity

scalar quantity [2]

- (b)** An aeroplane flies through the air in a straight line, with its engines at full power. The weight of the aeroplane is 160 kN. There is also a force of 320 kN acting upwards on the aeroplane at 30° to the horizontal.

In the space below, use a graphical method to determine the size and the direction of the resultant of these two forces. Use a scale of 1.0 cm : 40 kN.

size of resultant force =

direction of resultant force =

[4]

6. O/N/13/22/Q1

A set of traffic lights hangs from the end of a metal cable. A horizontal chain pulls the traffic lights to the right so that they are above the middle of the road. Fig. 1.1 shows the metal cable inclined to the vertical.

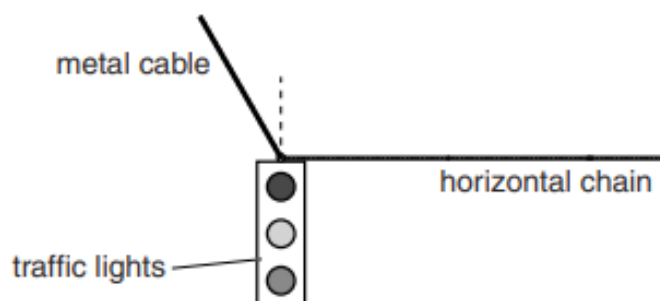


Fig. 1.1

The weight of the traffic lights is 240 N.

- (a) Two of the forces on the traffic lights are the tension in the horizontal chain and the weight of the traffic lights.

On Fig. 1.1, mark

- (i) an arrow that represents the tension in the horizontal chain, [1]
- (ii) an arrow that represents the weight of the traffic lights. [1]
- (b) The tension in the horizontal chain is 140 N. Use a scale diagram to determine the size of the resultant of the weight and the tension in the chain. State the scale used for the diagram.

scale =

resultant force =

[3]

7. O/N/12/21/Q1

A parachutist of total weight 950 N falls vertically at a constant velocity of 6.5 m/s.

- (a)** State the size and direction of the resistive (drag) force acting on the parachutist.

size of resistive force:

direction of resistive force:

[2]

- (b)** At a certain height, the wind starts to blow and the parachutist moves horizontally at a velocity of 4.5 m/s as he continues to fall.

Use a graphical method to add together the horizontal and the vertical velocities to determine the size of the resultant velocity of the parachutist. Label the velocities and state the scale that you use.

size of velocity =

scale =

[3]

8. M/J/10/22/Q1

A journey consists of two displacements: the first is 500m in a northerly direction and the second is 200m in an easterly direction.

(a) In the space below draw, to scale, a vector diagram of these displacements.

State the scale of your diagram.

On your diagram, show the two displacements and the resultant displacement.

Determine the size (magnitude) and direction of the resultant displacement.



scale =

size =

direction =

[3]

(b) Another journey covers a distance of 700m. Describe how it is possible that this journey has no resultant displacement.

.....

.....[1]

9. O/N/08/Q1

A microphone has a weight W of 6.0 N. It is suspended by wire X from the ceiling in a radio studio.

Fig. 1.1 shows the microphone held in the correct position by a horizontal wire Y.

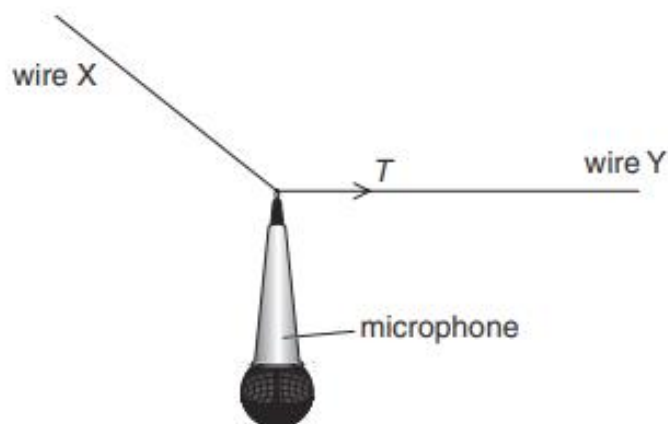


Fig. 1.1

- (a) The tension T in wire Y is 8.0 N. Use a vector diagram to determine the magnitude and the direction of the resultant of W and T .

magnitude =

direction =

[4]

- (b) The microphone is at rest. State the tension in wire X.

tension in X = [1]

ANSWERS

1.

Question	Answer	Marks
1(a)	(a vector quantity) / it has a direction	B1
1(b)	acceleration and momentum underlined and no others	B1
1(c)	speed: ($v = \sqrt{6.8^2 + 2.4^2}$ or $v^2 = 6.8^2 + 2.4^2$ or correct triangle / rectangle drawn / intersecting arcs (by eye)	B1
	speed: 7.1 – 7.3 (m / s)	B1
	angle: ($\theta = \tan^{-1}(6.8 / 2.4)$ or $\tan \theta = 6.8 / 2.4$ or 18 – 21(°) or correct triangle / rectangle drawn / intersecting arcs and scale <u>used</u> ≥ 1 cm: 1 m / s	B1
	angle: 69 – 72(°)	B1

2.

Question	Answer	Marks
7(a)(i)	85 m	B1
7(a)(ii)	(speed =) d / t in any form algebraic or numerical	C1
	2.1 m / s	A1
7(b)(i)	displacement involves direction or is a vector	B1
7(b)(ii)	5 m shown on dotted line	B1
7(b)(iii)	length of diagonal 9.2 (cm) $\sqrt{30 \times 30 + 35 \times 35}$	B1
	46 m 46 m	B1
7(b)(iv)	39–42° or 318–320°	B1

3.

Question	Answer	Marks
1(a)	scale vector diagram showing 12 N, 36 N and correct resultant force in correct direction	B1
	38 ± 1 kN	B1
	$018 \pm 3^\circ$	B1

4.

Question	Answer	Marks
1(a)(i)	vector quantity has direction / scalar quantity does not have direction	B1
1(a)(ii)	any two scalar quantities, e.g. mass, distance / length, time, speed	B1
	any two vector quantities, e.g. velocity, acceleration, displacement, force / weight	B1
1(b)	P and Q drawn in correct directions to meet at a point with correct size	B1
	triangle or parallelogram shown with all three vectors in correct directions and the resultant clear (e.g. as another vector)	B1

- 5.
- (a) (i) (vector) has direction **or** scalar does not have direction **or**
(vectors) may cancel **or** scalars cannot cancel B1
- (ii) one vector quantity e.g.: displacement; velocity, acceleration B1
one scalar quantity e.g.: distance, length, speed, time, current, resistance B1 [3]
- (b) downward weight arrow of length 3.9–4.1 cm B1
correct triangle/parallelogram drawn **and** correct diagonal clear B1
270–285 kN B1
horizontal ($\pm 3^\circ$) B1
- 6.
- (a) (i) arrow(head) on chain pointing to the right B1
- (ii) vertical arrow downwards and part of arrow touching **or** within rectangle
of lights **or** direction of arrow in (i) **and** (ii) correct (by eye) B1
- (b) scale given (must have unit of cm:N **or** cm/N **or** N:cm **or** N/cm) B1
correct triangle **or** rectangle (might be implied) and correct resultant
(compulsory e.c.f. from (i) **or** (ii): i.e. correct diagonal according to
candidate's (i) and (ii)) B1
 $272 \leq \text{candidate's value} \leq 283 \text{ N}$ B1 [5]
- 7.
- (a) 950 N B1
upwards B1
- (b) correct rectangle **and** diagonal **and** at least one velocity labelled
or correct triangle and at least one velocity labelled
(either way round) B1
from 7.8(0000) to 8.0(0000) m/s (inclusive) B1
scale stated B1 [5]
- 8.
- (a) components shown on correct diagram with correct resultant (i.e. towards NE) and
a scale given B1
540 (± 10) m B1
 $22^\circ \pm 3^\circ$ E of N with correct diagonal B1 [3]
- (b) idea that ends at start, returns in opposite direction B1 [1]
- 9.
- (a) diagram of two forces **and** resultant B1
W / 6(N) **and** T / 8(N) marked on perp. forces **or** scale given B1
10(.0 ± 0.2) N B1
35–39° from T/Y/horizontal or 51–55° from W/vertical **and** correct resultant B1
- (b) 10(.0) N **or** e.c.f. B1 [5]